Zia Choudhry, MD, PhD, MBA

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Pharmaceutical Medical Director with Early/Late-Phase Development Experience and Medical Affairs Expertise focusing on Multiple Disease Areas and Therapeutic Landscapes in Global, US and Emerging Markets

Executive Profile

Dr. Zia Choudhry is a pharmaceutical executive, medical affairs strategist, clinical research development professional, medical pharmacovigilance specialist, personalized health expert, and medical geneticist. Dr. Choudhry has 17+ years of experience in clinical care, clinical trials research, medical affairs, and pharmacovigilance, during which he has held various leadership positions and led teams for top tier organizations, including Purdue, Genomind, and Pfizer. He has led Biopharmaceutical, PGx, and medical device pivotal clinical trials (Phases I to IV), product launches, and communication platforms. He has published in prestigious scientific journals, conducts CMEs for physicians, and serves on the boards of two non-profit organizations. Dr. Choudhry received Medical Degree from University of Karachi and pursued an Internal/Family Medicine residency. He also holds a Master of Health Informatics from SZAB Institute of Science & Technology; a Master of Business Administration from Al-Khair University; and a Master of Sciences in Psychiatry, a Post Graduate Certificate in Sleep Medicine, as well as a Doctorate of Philosophy in Human Genetics from McGill University, Canada. Dr. Choudhry is committed to development of Drugs, Devices and Diagnostics that help improve patients' lives.

Medical and Scientific Specialties

- Clinical Development, Medical Affairs & Business Development
- Global and US Research and Drug Development Strategy and Tactical Leadership (Medical monitoring, Therapeutic Area Subject Matter Expertise, Project/Program Feasibility Evaluation, Development/Review of Protocols, Supervision of Clinical Studies Conduct, Adherence of GCP and SOPs in Clinical Trials, Clinical Training of Internal and External Stake-holders, Expert Investigators Evaluation and Relationship Management, Medical leadership for SAEs and narrative summaries, Clinical operations collaborations, Presentations related to trials/results, Fielding of Medical queries, Conference participation and engagements, etc.)
- Global and US Medical Strategy and Tactical Leadership (Benefit-Risk Optimization for Brand success, IMEs, Symposiums, Advisory Boards, KOL Management, Publication Planning, Patient Journey, Scientific Communications Plans, Insights generations, Expanded Access Program, Patient and Professional Organization Relationships, etc.)
- Family and Internal Medicine Training
- Clinical Neurosciences, Rare Diseases & Oncology Clinical Experience
- Medical management and monitoring of global/regional clinical trials (Phases I – IV), Registries, RWE & HEOR Studies.
- Pharmacovigilance: Patient Safety, Risk Management & ICSR, SER, Periodic Reports (PBRER, DSUR, ACO) Review
- Safety Governance Meetings (SMT & SMC)
- Biomarker Research & Discovery
- Epidemiology, Big Data & Predictive Analytics
- Personalized & Digital Medicine
- Biopharma/Med-Device/Med-Diagnostics Expertise
- Pharmaceutical Strategy, Market Access & Commercialization
- Medical Writing, Clinical training, CME & Grants

- **Internal & Family Medicine:** Hypertension, CAD, MI, CHF, CVA, Hyperlipidemia, Diabetes Mellitus, Obesity, Thyroid Disorders, Thyroid Cancer, Addison's Disease, Cushing's Syndrome, Osteoporosis, Rheumatoid Arthritis, Inflammatory Disorders, Kidney Transplantation, GvHD, and other medical disorders
- **Neurology:** Alzheimer Disease, Dementia, Multiple Sclerosis, Neuromyelitis Optica Spectrum Disorder, Myasthenia Gravis, Parkinson Disease, Epilepsy and Seizure Disorders, ALS, Huntington Disease, Spinal Muscular Atrophy, Neurogenic Orthostatic Hypotension, Tauopathies, Synucleinopathies, and other neurological disorders
- **Psychiatry:** Major Depressive Disorder, Bipolar Disorder, Agitation, Cognitive Dysfunction, ADHD, Tourette Syndrome, Sleep Disorders, Anxiety Disorders, OCD, Post-Traumatic Stress Disorder, Borderline Personality Disorder, Eating disorders, Schizophrenia, and other psychiatric disorders
- **Pain Medicine:** Low Back Pain, Chronic Pain disorders, Cancer Pain, Post-surgical Pain, Arthritis, Headache, Migraine, Fibromyalgia, Neuropathic pain and other pain disorders
- **Rare Medicine:** Lysosomal Storage Disorders, Acid Sphingomyelinase Deficiency, Fabry disease, Pompe disease, CARASIL, Velocardiofacial syndrome, and other rare disorders
- **Oncology:** B-cell chronic lymphocytic leukemia (B-CLL), B cell malignancies, non-Hodgkin's lymphoma (NHL), Diffuse large B cell lymphoma (DLBCL), Multiple Myeloma, and other oncological disorders

Career Summary and Highlights

- **Extensive Medical Affairs and R&D Management experience** in global pharmaceutical, start-up biotechnology and clinical research organizations
- Accomplished **pharmaceutical physician**, **strategist**, **business development professional** and **project manager** with diversity of experience in multiple therapeutic areas in complex multicultural **matrix environments**
- Experienced in tackling complex and complicated pharmaceutical challenges by employing **scientist** & **physician qualifications**, along with **management skills** and the ability to transform **Medical Affairs** function into a key **strategic business differentiator** and **key contributor** to the success of the organization
- In-depth understanding of both OTC & Prescription Markets, expertise in leading **cross-functional** teams for evaluation & development of new assets (**early-stage**), and proficiency in providing medical oversight and leadership on all major medical, marketing and regulatory activities for the inline products (**launch & late-stage**)
- Extensive expertise in developing **Medical Affairs strategies, plans and budgets** and proven ability to execute efficiently, strategically and on budget to achieve strategic imperatives
- Leadership capabilities in **cross-functional matrix environments** and the ability to **develop strong relationships** and to **influence team-members** in achieving consensus and in making **strategic decisions**
- Strategically use complex data and develop communication platforms to advance knowledge and product awareness
- Experienced in **Pharmacovigilance**: medical review of safety documents (including **PBRER, DSUR, SERs**) **Risk Assessment/Risk Management/ Benefit-Risk Assessment**, including but not limited to: provide proactive risk assessment. Co-lead benefit-risk assessment with other relevant functions, develop risk management strategies and plans and monitor effectiveness
- Review **safety regulatory and clinical documents** including submission, transversal and primary GPE documents
- **Personalized Medicine, Internal Medicine, CNS, Oncology and Rare Genetic disorders KOL** with **50+ research publications** in high- impact scientific journals
- Veteran **clinical investigator** with experience in leading **20+ clinical research programs** (**POC to phases I- IV**) in children, adolescents, adults, geriatrics and additional specialized populations
- Proficient member of the **Medical Safety Committee** trained in evaluating and reporting **adverse events** and other safety information relevant to new & mature products for local & global markets
- **Knowledgeable scientific expert & partner** to Regulatory Affairs, Drug Safety, Managed Markets, Commercial, and Health Policy leads in preparing reports & presentations for HCPs, Payers & additional external stakeholders

Employment and Experience

2018 - Present **Sanofi Genzyme.**
Global Medical Director, Medical Affairs – CNS / MS Franchise (Global Medical Affairs)

Medical Affairs Strategy & Tactical Leadership:

- Provide medical leadership to the development and implementation of the brand strategy, taking into account the Benefit/Risk profile of the drug to optimize brand success in a highly competitive CNS environment

- Define and develop KOL management plan with top Global and US KOLs, ensuring recommendations are acknowledged in Sanofi plans; developing best brand knowledge with top KOLs to ensure accurate communication
- Partner with clinical, pharmacovigilance, statistics teams to ensure best post-hoc analyses of the data, Benefit/Risk profile, target population, to build foundation and differentiation of the brand
- Develop and execute communication strategy and scientific platforms in collaboration with KOL Pub Committees, internal contributors, Communication Director, and agencies
- Lead cross planning, and actively participate in the execution of brand communications plan supporting scientific communications needs and core strategy (e.g. medical society presence, HCP interactions, abstracts and publications development, etc.)
- Lead liaising efforts with Regional Medical Directors and US Medical Brand Leads to ensure fluid scientific communication, satisfy medical needs to help in medical education, market access and commercial success
- Define timelines, budgets, and tactical plans and establish a strong network among all team members in order to effectively recommend and influence key strategic business decisions
- Document, coordinate, and review Periodic reports, e.g.: RMP update, Annual Safety Report/Development Safety Update Report, Periodic Safety Update Report
- Successfully managed and coordinated Medical, Safety, Clinical and Commercial activities during Article 20 procedure for Lemtrada, resulting from the evaluation of adverse event data from pharmacovigilance activities. Facilitated strategic thinking and alignment, engaged KOLs and reviewed AEs of special interest, conducted Advisory board meetings and gathered expert insights and feedbacks, developed deliverables (reports, presentations, etc.), and conducted high-impact labeling discussions with EMA to achieve favorable outcome
- Provide medical leadership to the development and implementation of the brand strategy of oncology and other indications for Alemtuzumab, taking into account the Benefit/Risk profile of the drug to optimize brand success and patient care

Clinical Development/Operations Strategy & Tactical Leadership:

- Successfully managed assigned global and regional clinical trials (Phases IV) for CNS products and served as Global Medical Lead to collaborate with the other cross-functional key members including pharmacovigilance team
- Acted as the Subject Matter Expert and provided therapeutic and protocol-specific key insights and knowledge to support internal and external training and development of project team members
- Provided Scientific Expertise and helped by sharing medical and clinical input into the effective design of study protocols, clinical development program plans, research papers, and other associated deliverables
- Supported Medical Monitoring activities for all assigned clinical projects and supported other CNS studies as needed to facilitate timelines and business imperatives
- Provided 24/7 Medical Expertise to all regional and global assigned clinical projects to ensure timely query resolution
- Supported Drug development activities (New indications, Biomarkers, etc.) as a key member and provided medical and clinical insights and knowledge to facilitate protocol consideration/review, feasibility assessments, review of proposals/contracts for medical services, and attendance at meetings with CROs. Also supported in the review process for final study reports (CSRs), or other study documentation (Protocols, ISS/ISEs etc.) created by CROs and other vendors
- Support and mentor other team members and consistently displayed exemplary work ethics, compassion and integrity, to support organizational strategic imperatives and goals

2017 - 2018 **Purdue Pharma L.P.**
Medical Director, Medical Affairs Products, Pipeline & Strategy – Pain Medicine / CNS Franchise and Pipeline Oncology Assets (US and North American Medical Affairs)

- Provided medical expertise & direction for assets in development with regulatory interactions & submissions
- Developed and managed medical, disease area, and organizational strategic plans in collaboration with R&D, commercial, BD, safety, regulatory and additional cross-functional colleagues
- Established and managed the "Medical Clinical Product team" (9 professionals representing MA and clinical sub-functions) and provided strategic planning, coordination & execution of medical-scientific communication, evidence generation, medical education and research activities related to pain product portfolio and pipeline oncology assets (tinostamustine, and bendamustine)

- Engaged with top tier international/national KOLs in coordination with MSLs, health-policy, government affairs, managed markets and commercial colleagues in order to recommend and influence key strategic business decisions
- Provided medical clinical management for all clinical trial projects (Phases I – IV) as the Medical Global and US Clinical Development Lead Medical Monitor for pain medicine and for inline and pipeline products
- Collaborated with cross-functional key members including pharmacovigilance team to efficiently process SAEs and SUSARs
- Provided medical review of safety documents (including PBRER, DSUR, SERs) and led Risk Assessment/Risk Management/ Benefit-Risk Assessments, developed risk management strategies and plans and monitor effectiveness
- Review safety regulatory and clinical documents including submission, transversal and primary GPE documents
- Responsible for 24/7 Medical Expertise to all regional and global pain medicine and pipeline oncology clinical projects to ensure timely query resolution including questions pertaining to inclusion and exclusion criterion, lab measures, and safety aspects of the study
- Led all clinical trials related activities requiring Medical & Scientific input, including study protocol development and review, feasibility assessments, review of proposals/contracts for medical services, and attendance at meetings with CROs. Also supported in the review process for final study reports (CSRs), or other study documentation (Protocols, ISS/ISEs etc.) created by CROs and other vendors
- Provided therapeutic area training and mentored other team members, also responsible for providing clinical input to the design of study protocols and clinical development programs to support organizational strategic imperatives and goals

2015 - 2017 **Genomind Inc.**
Chief Medical Officer – Senior Vice-President Medical Affairs and Clinical Development (Global and US Markets)

- Led a team of 8 professionals (5 MScs and 3 PhDs) and effectively conceived & launched clinical trials, medical scientific communication platform which increased HCP & payer confidence in Genomind products translating into an increased loyal HCP prescriber base and reimbursement wins
- Headed R&D efforts for novel DGx and PGx (Interventional clinical trials - NCT02566057 and NCT02634177) product development and led medical affairs strategic research projects (resource utilization evidence creation) for HCPs, Payers and other stakeholders
- Responsible for planning and leading all Clinical Development activities as the CMO and provided medical and clinical insights and knowledge to facilitate proposal review, protocol consideration/review, feasibility assessments, review of proposals/contracts for medical services, and attendance at meetings with CROs
- Led the review process for final study reports (CSRs), and other study documentation (Protocols, ISS/ISEs etc.) created by CROs and other vendors
- Efficiently developed medical affairs strategies and tactics for supporting business objectives including, KOL engagement, evidence generation platform, and investigator sponsored trials, publication planning, CME grants/sponsorships, speaker bureau, medical advisory boards, and commercial promotional review
- Effectively executed Scientific Advisory Boards and developed strong relationships with KOL networks gaining impactful expert insights influencing product life cycle planning which supported business, clinical and scientific goals

2014 - 2015 **Premier Psychiatric Group, LLC.**
Medical Director – Medical Affairs and Medical Monitoring*, Premier Psychiatry Research, LLC*
Medical Director – Medical Affairs – Transcranial Magnetic Stimulation (TMS) medical device franchise*, Nvation, LLC & Brainsway, LTD*

US and North American Medical Affairs & Medical Monitoring Lead:

- Served as the Lead Medical Monitor and provided 24/7 medical oversight for global clinical trials (Phases I – IV), responsible for managing a team of clinical research medical professionals (4 MDs) and for leading medical clinical management services for all assigned drugs, devices and biologics studies

- Collaborated with key members of the Medical & Safety project teams to ensure timely evaluation and assessments, and for effective follow-up and processing of AEs, SAEs, AESIs
- Responsible for leading development of clinical development plans, protocol design, and risk assessments for all CNS projects
- Provided medical leadership to clinical trial sites by continued investigator engagement, study protocol training, review/analysis of safety/efficacy trends, and the medical management of subject emergencies
- Proactively supported study team with scientific medical and clinical questions/queries including, subject eligibility, protocol related medical specific question, assessment of exclusionary and alert laboratory values
- Responsible for Coding and review of the protocol deviation/violation, and for proactively identifying clinical issues requiring MM actions: queries, tracks, and follow resolution to the trackers
- Served as lead medical director and assisted in the preparation and review of final narratives, clinical study reports (CSRs), and other study documentation (Protocols, ISS/ISEs etc.)
- Supported Business Development activities as the scientific lead and provided medical knowledge expertise to facilitate proposal generation, protocol consideration, feasibility assessments, review of proposals/contracts for medical services, and attendance at meetings with sponsors
- Developed and maintained an expert level of scientific competence related to assigned clinical trails, including therapeutic area knowledge, marketplace developments, competitor landscape, project related information and led external/internal meetings for key medical questions and clinical queries related to study
- Maintained good knowledge of GCPs and all regulatory policies and procedures (FDA, EMA, emerging markets) relating to clinical development and safety to ensure compliance with ethical, legal and regulatory standards.
- Mentored and developed reporting medical team members inline with organizational development objectives and assisted in the attainment of annual departmental revenue targets

US and North American Medical Affairs Lead - TMS medical device franchise:

- Led a team of medical affairs professionals (3 PhDs) and developed cross-functionally aligned medical strategic and tactical plans/budgets, scientific communication platforms, product monographs & medical materials, CME programs
- Conceived, planned and initiated strategic KOLs, payers and policy stake-holders engagement programs influencing earlier adoption of TMS therapy in Substance Abuse, MDD and other neurological and psychiatric disorders
- Presented pivotal trial data and outcomes research findings at the National Clinical TMS Society & ISEN meetings 2014/2015, and advocated TMS appropriate utilization and access at the 2014/2015 APA meetings

2004 - 2013 **Douglas Mental Health University Institute (DMHUI) & Departments of Psychiatry and Human Genetics, McGill University.**
Medical Advisor & Clinical Research Physician Scientist – Genetics/ Clinical Neurosciences/ Oncology Clinical Trials

- Co-Principal/Sub Investigator (Co-PI/Sub-I) on several academic research grants, observational & surveillance studies, Phase I – IV clinical trials sponsored by biopharmaceutical industry. Provided medical strategic expertise and scientific tactical leadership to the grantsmanship process, budget formulation, project management, contract review, and stakeholder liaising for multi-million-dollar projects
- Effectively led collaboration with >20 academic groups and multiple biopharmaceutical, medical device and diagnostic companies to advance experimental drug, device and diagnostic research resulting in pivotal interventional and non-interventional evidence creation for multiple innovative healthcare products, resulting in high impact publications and regulatory reports
- Successfully led BD efforts for scientific scholarly research and data collection/management services, helped industrial partners in developing new viable products by establishing research partnership agreements
- Supervised BSc, MSc, MD level research team engaged in multiple clinical trials (Phases I to IV)

2001 – 2004 **Pfizer Pharmaceuticals.**
Medical Affairs Lead – CVS and CNS Products Portfolio (Emerging Market Medical Affairs)

Medical Affairs Lead - CVS & Primary Care Portfolio:

- Conceived, created & implemented medical and clinical strategic and tactical Plans for the successful registration and launch of CAD & ISH indications for **Norvasc** (late stage), launch of **Lipitor,** launch of **Zithromax,** Launch of **Cardura** (BPH and high blood pressure indications)
- Provided Medical leadership for emerging markets and facilitated Global Phase II/IIIb/IV studies, Investigator Sponsored Studies and other R& D studies in close collaboration with the Global R& D
- Developed and communicated the emerging markets scientific publication plans, provided oversight of all publications including posters, abstracts, manuscripts at the regional level to ensure appropriate alignment between relevant teams
- Developed regional RWE/HEOR plans, supervised on-going data generation projects as well as provided guidance for new projects
- Led frequent communication with the regional MSL teams via the MSL Lead, ensured appropriate and timely training on new data, supported answering medical clinical questions from the team
- Acted as Emerging Market Medical Clinical liaison with the Global Medical and Clinical teams to ensure appropriate strategic alignment in the planning process
- Facilitated the grants and scientific review committees, ensured timely reviews of proposals and made decisions based on scientific strategic imperatives
- Developed and maintained excellent relationships, ensured accurate communication with appropriate external KOLs and experts in the CVS and Primary Care therapeutic areas in the emerging markets
- Supported the evaluation of any business development opportunities
- Provided budget oversight and compliance with all regional and local laws, regulations and policies that governed the conduct of Medical Affairs and Clinical Research activities

Medical Affairs Lead - CNS Portfolio:

- Coordinated and developed the Emerging Markets Medical Plans in collaboration with cross-functional teams to support international (non-interventional/interventional) evidence creation and dissemination, KOL development facilitating "Therapeutic Area Disease Awareness" and launch of **Neurontin, Lyrica, Zoloft (MDD)**

Education and Certifications

2017 - 2018	**Postgraduate Certificate – Biomedical and Health Informatics (PgCert);** Oregon Health & Science University
2008 - 2013	**Doctor of Philosophy – Human Genetics (PhD);** McGill University & DMHUI
2007 – 2008	**Postgraduate Certificate – Sleep Medicine (PgCert);** Michener Institute
2005 – 2007	**Masters of Sciences - Psychiatry (MSc);** McGill University, & DMHUI
2001 – 2003	**Masters of Business Administration (MBA);** Al-Khair University
2000 – 2002	**Masters of Computer Sciences – Health Informatics (MCS);** SZAB Institute of Science & Technology
1997 – 2001	**Postgraduate Medical training – Family/Internal Medicine;** JPMC & Mumtaz Hospital
1992 - 1997	**Bachelor of Medicine and Bachelor of Surgery (MBBS/MD equivalent);** University of Karachi

Medical Affairs Achievements and Accomplishment

- Successfully partnered with cross functional teams and conducted 4 **Advisory Boards (strategic, international, safety, research, EU and US) for Lemtrada**, engaged KOLs, gathered impactful clinical and strategic insights influencing product life cycle planning which supported business, clinical and scientific strategic imperatives and goals
- Achieved operational excellence as the Medical Affairs leader by providing scientific strategic leadership for Medical Affairs functions in **pain franchise (OxyContin, Hysingla ER and all inline & non-promoted products) and oncology pipeline**
- Successfully designed and implemented **Personalized Medicine, CNS (psychiatry, neurology and pain) and genetic disorders** therapeutic area strategies, efficiently launched multiple products and product line extensions, and developed international business partnerships
- Accomplished profitable growth by facilitating CEO in building veteran scientifically trained medical affairs, clinical research and commercial teams

- Attained distinction as the organizational Medical affairs leader by assisting SVP Molecular Pathology in building a genetic testing laboratory accredited by CLIA and CAP and licensed in all 50 states and transitioning 100% of Genetic testing volume to internal laboratory
- Successfully led clinical evidence development efforts by conceiving, designing, managing and completing 9 clinical research projects, including two pivotal clinical trials – one with a major national insurance carrier and the other Phase III Double Blind Multicenter study (24 clinical sites across the US) (NCT02634177)

Detailed related to professional experience, roles and responsibilities, strategic medical affairs and R&D projects, publications and presentations, professional memberships and references available on request.